

DAYTONA BEACH, FL, USA

WMO: 747870

Lat: 29.183N

Lon: 81.048W

Elev: 10

StdP: 101.21

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12834

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	4.2	-6.5	2.2	6.7	-3.4	2.8	8.1	10.4	16.4	9.1	15.4	2.5	320	0.358

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	33.7	25.1	32.7	25.1	31.9	25.0	26.9	30.9	26.4	30.5	26.0	29.9	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	28.6	25.4	20.6	28.2	25.0	20.1	28.0	84.1	31.0	81.9	30.4	80.3	30.0	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.0	7.2	-1.1	35.6	1.9	1.0	-2.5	36.3	-3.6	36.9	-4.6	37.5	-6.0	38.2		
			-3.1	27.6	2.0	0.9	-4.5	28.3	-5.6	28.8	-6.7	29.2	-8.2	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	14.7	16.3	18.4	21.2	24.3	26.7	27.6	27.7	26.7	23.6	19.4	16.7
	DBStd	5.56	4.75	4.44	3.93	3.00	2.22	1.47	1.10	1.08	1.31	3.02	3.77	4.66
	HDD10.0	27	14	6	1	0	0	0	0	0	0	0	1	6
	HDD18.3	394	129	82	48	11	1	0	0	0	0	4	33	86
	CDD10.0	4396	161	183	262	337	443	502	544	547	500	423	281	214
	CDD18.3	1719	17	26	50	97	184	252	286	289	250	168	64	37
	CDH23.3	14361	73	135	284	646	1502	2226	2758	2836	2242	1227	294	138
CDH26.7	4688	3	16	49	136	467	829	1099	1088	706	252	33	10	
Wind	WSAvg	3.0	3.1	3.2	3.4	3.3	3.1	2.6	2.3	2.4	2.9	3.2	3.0	2.9
Precipitation	PrecAvg	1233	67	80	90	53	76	160	148	158	168	104	71	61
	PrecMax	1879	182	232	268	157	313	386	366	294	409	296	278	304
	PrecMin	848	8	10	8	0	2	21	4	51	11	5	0	5
	PrecStd	250	48	55	69	43	64	94	79	68	89	70	64	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	28.5	30.0	31.4	34.0	35.3	34.7	34.3	33.0	31.4	29.2	27.9
		MCWB	20.6	20.9	20.8	21.6	22.7	25.1	25.7	25.5	24.7	24.5	22.9	22.2
	2%	DB	25.4	26.8	28.0	29.7	32.1	33.3	33.4	33.1	31.8	30.1	27.6	26.3
		MCWB	19.9	20.1	20.4	21.6	22.6	24.9	25.6	25.6	24.9	24.3	22.2	21.4
	5%	DB	23.7	25.1	26.4	28.1	30.6	32.0	32.4	32.3	30.9	29.0	26.3	24.9
		MCWB	19.1	19.6	19.8	21.0	22.8	24.9	25.6	25.6	24.9	23.7	21.3	20.6
	10%	DB	22.1	23.2	24.7	26.9	29.2	30.9	31.4	31.3	30.1	28.1	25.0	23.4
		MCWB	18.3	18.6	19.0	20.5	22.5	24.6	25.4	25.5	24.7	23.1	20.5	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.5	22.9	24.1	25.3	27.0	27.4	27.7	26.9	26.5	24.7	23.4
		MCDB	25.0	25.7	26.7	27.9	29.8	31.6	31.9	31.6	30.4	28.7	26.7	26.2
	2%	WB	20.9	21.5	21.9	23.2	24.6	26.3	26.8	26.9	26.3	25.6	23.8	22.4
		MCDB	23.6	24.6	25.7	26.8	28.9	30.9	31.2	30.8	29.6	28.3	26.0	24.8
	5%	WB	20.0	20.6	20.9	22.4	24.1	25.7	26.3	26.4	25.8	24.9	22.7	21.5
		MCDB	22.7	23.7	24.6	26.1	28.1	30.0	30.7	30.4	29.1	27.8	25.0	23.9
	10%	WB	19.0	19.6	20.1	21.6	23.5	25.2	25.8	25.9	25.4	24.3	21.6	20.6
		MCDB	21.6	22.5	23.5	25.4	27.4	29.3	30.0	29.7	28.7	27.0	24.1	22.9
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.7	10.6	10.4	9.8	8.7	8.6	8.1	7.3	8.1	9.5	9.9
		MCDBR	11.2	11.6	11.8	11.5	11.8	10.4	9.7	9.3	8.3	8.0	9.5	9.6
		MCWBR	6.3	5.9	5.4	4.9	4.3	3.9	3.7	3.6	3.2	3.5	4.9	5.3
	5% WB	MCDBR	9.9	10.1	10.1	9.9	9.5	9.2	8.9	8.2	7.2	6.7	7.8	8.6
		MCWBR	6.1	5.8	5.3	4.9	4.1	3.9	3.7	3.6	3.3	3.2	4.7	5.3
Clear-Sky Solar Irradiance	taub	0.341	0.354	0.372	0.394	0.429	0.458	0.483	0.471	0.425	0.396	0.360	0.348	
	taud	2.539	2.510	2.457	2.375	2.321	2.311	2.242	2.284	2.408	2.459	2.505	2.538	
	Ebn at Noon	899	915	917	901	866	836	814	822	853	858	868	872	
	Edh at Noon	88	98	110	123	130	131	140	133	114	102	90	85	
All-Sky Solar Radiation	RadAvg	3.25	3.92	5.08	6.14	6.42	5.96	5.94	5.62	4.89	4.33	3.49	2.97	
	RadStd	0.24	0.32	0.40	0.33	0.43	0.55	0.34	0.37	0.35	0.32	0.20	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.40	N/A	N/A	N/A	N/A	N/A	-8	-48	+136	+112
(47) Regional (37 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-6	-50	+152	+110

Nomenclature: See separate page